

Accops Solutions for Government



Introduction

This document describes the proposed security products and reference architecture designed to strengthen the IT infrastructure of Government Organisations. The proposed solutions follow a **Zero Trust security model** with strong identity verification, secure access controls, endpoint isolation, and centralized management.

It consists of the following integrated / 'off the shelf' components:

1. Accops ZTNA for Secure Private Application Access
2. Accops HyID (MFA, SSO, IAM)
3. Accops VDI Suite (VDI + ZTNA + MFA)
4. Accops Remote Browser Isolation (Internet Isolation)
5. Accops Airbridge (Secure File Transfer Gateway)
6. Accops Huddle (Secure Remote Support for IT Operations)
7. Accops Thin Client – HyDesk (Secure Endpoint Computing)

Accops ZTNA – Secure Private Application Access

Product Overview

Accops ZTNA provides secure Zero Trust Network Access to internal enterprise applications hosted within government data centers. The solution replaces traditional VPN-based access with identity-based and application-level access control. Users are granted access only to specific applications after verifying identity, device posture, and contextual parameters such as location and time.

Key Capabilities

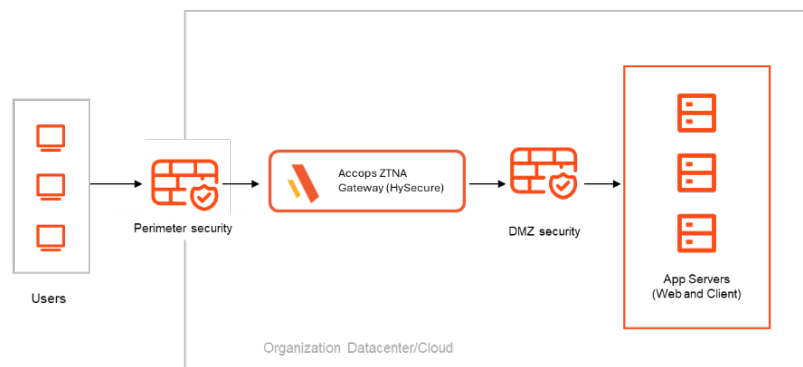
- Deliver Secure access to web applications, RDP, SSH and internal applications
- Layer 7 reverse proxy for application access
- Layer 4 application tunnelling
- Device posture validation and endpoint compliance checks
- Context-based access policies
- Support for external QRNG for PQC

- Enterprise browser with granular controls
- Integration with LDAP, Active Directory, RADIUS, and SAML. Support for Native Authentication
- Clientless browser-based access as well as agent-based connectivity
- Data leakage prevention controls such as blocking copy-paste, screenshots, and screen recording
- Web Application Protection against threats such as OWASP Top 10, OWASP Top 20 automated threats, and SANS Top 25 vulnerabilities.
- Secure access for internal, internet, vendors and third-party users

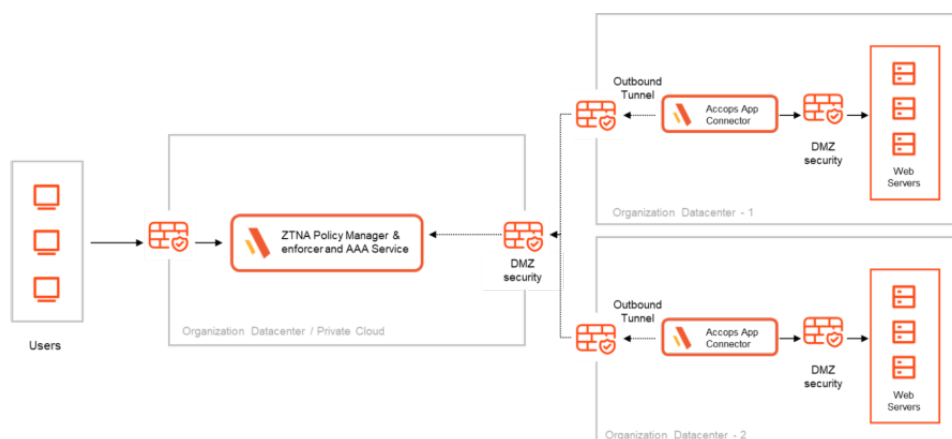
Reference Architecture

Deployment Options:

- **On-Premises – Direct Connect:** The solution is hosted in your data center, and external users connect directly through inbound firewall access.
- **Cloud / As-a-Service – Direct Connect:** The solution is cloud-hosted, and users connect directly to it over the internet.



- **On-Premises – Reverse Connect (with Edge Connector and App Connector):** The solution is deployed on-premises and uses outbound-initiated tunnels via Edge and App Connectors to securely enable external access without opening inbound ports.
- **Cloud / As-a-Service – Reverse Connect (with Edge Connector and App Connector):** The cloud service uses outbound-initiated tunnels through Edge and App Connectors from your environment to provide secure access without inbound firewall exposure.



Flexible Model:

- **Self-Hosted and Managed (Cloud or On-Premises):** The solution is deployed in your environment (cloud or on-prem) and fully controlled and managed by your team.
- **ZTNA as a Service (Cloud or On-Premises):** The solution is delivered and managed as a service by the vendor, whether hosted in the cloud or deployed on-premises.

Use Cases

- **Securely Connect to All Corporate Resources:** Enable employees to access any corporate application—including web, client-server, RDP, SSH, and file servers—from any location without exposing the internal network. This access is reinforced by WAP's comprehensive security framework, which proactively shields web applications and APIs from a wide range of sophisticated cyber threats.
- **Provide Universal, Secure Access for All User Types:** Extend secure access to internal employees, contractors, vendors, and BYOD users. Guarantee the same protected experience regardless of the device (mobile, laptop, or desktop) being used.
- **Provide Agent-less Access for Field Workforces:** Enable secure, browser-based access to web applications for your entire field workforce and external partners. Eliminate the need for endpoint agent installation for seamless connectivity.
- **Rapidly Onboard and Scale Remote Access:** Instantly provision on-demand, role-based remote access for any number of users. Simplify user setup to ensure your workforce can begin working securely from anywhere, immediately.
- **Replace Legacy VPN Solutions:** Migrate from outdated, flat-network VPN architectures to modern ZTNA, reducing the attack surface and enabling contextual, least-privilege access across all users and devices.
- **Enable On-Demand Data Protection (DLP):** Implement immediate, contextual DLP controls and on-demand file encryption for sensitive data being transferred or accessed, ensuring critical information remains secure on unmanaged devices.
- **Consolidate Legacy Access Tools:** Replace multiple, fragmented legacy security appliances—such as VPNs, traditional Network Access Control (NAC), traditional Web Application Firewall (WAF) and client-side access portals with a single, unified software-based platform.

Accops HyID – Identity, MFA and Access Management

Product Overview

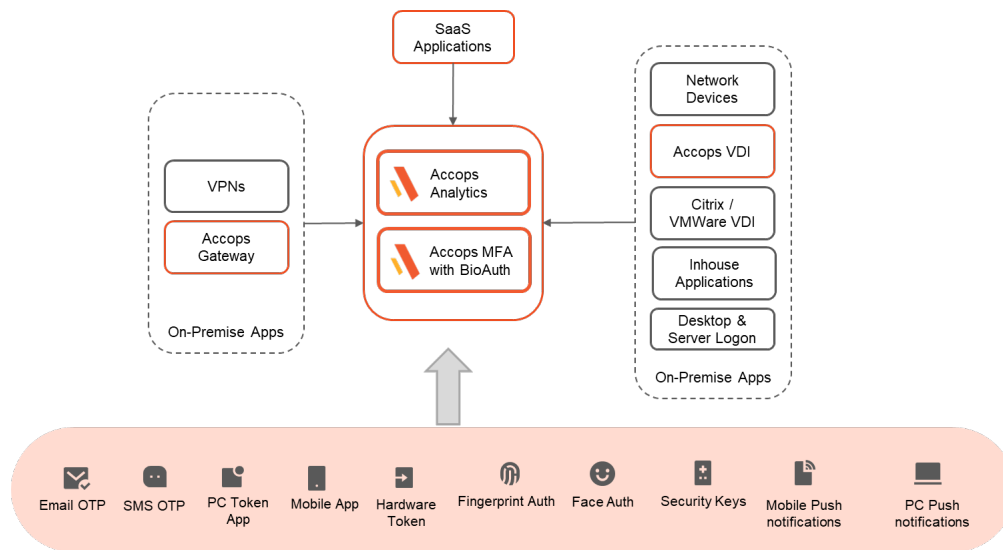
Accops HyID provides a comprehensive **Identity and Access Management (IAM) platform** with integrated **Multi-Factor Authentication (MFA)** and **Single Sign-On (SSO)** capabilities.

The solution supports **passwordless authentication, contextual access control, and integration with legacy and modern applications.**

Key Capabilities

- Multi-factor authentication using OTP, push notifications, biometrics, or hardware tokens
- QR Based Login - Passwordless authentication
- Integration with FIDO and Windows Hello
- Single Sign-On across enterprise applications including web, cloud, and legacy applications
- SSO support using industry-standard protocols such as SAML 2.0, OAuth
- Connectors for RADIUS based applications
- Context-based authentication policies based on user role, device posture, location, and time of access
- Integration with enterprise directories such as Active Directory and LDAP
- Secure login for desktops, servers, VPN, and web applications
- Impossible travel detection and device binding to prevent credential misuse
- Adaptive risk-based authentication based on user behavior and access patterns
- Biometric authentication support including fingerprint and facial recognition
- Integration with third-party identity providers and enterprise IAM systems
- Comprehensive authentication logging, monitoring, and SIEM integration for security auditing and compliance

Reference Architecture



Use Cases

- **Secure authentication for enterprise applications** enables users to access critical business applications through strong multi-factor authentication, ensuring that access is granted only after validating user identity through multiple authentication factors.
- **Single Sign-On for enterprise and government applications** allows users to authenticate once and securely access multiple applications including web portals, internal enterprise systems, cloud services, and legacy applications without repeated logins, improving user productivity and security.
- **Secure administrator authentication** provides strong identity verification for system administrators accessing servers, network devices, databases, and management consoles, ensuring privileged access is protected with multi-factor authentication and detailed audit logging.
- **Identity protection against credential misuse** includes mechanisms such as impossible travel detection, device binding, and risk-based authentication to prevent unauthorized account usage and identity-based attacks.
- **Unified authentication across infrastructure systems** allows secure login across desktops, servers, web applications, and enterprise platforms through a single identity framework integrated with Active Directory and LDAP directories.
- **Compliance and audit** requirements, it enables organizations to maintain detailed authentication logs, session records, and identity verification events, helping meet regulatory requirements and providing full visibility into user authentication activities

Accops VDI Suite – ZTNA, VDI & MFA

Product Overview

Accops VDI Suite - Digital Workspace provides a **virtual desktop infrastructure (VDI)** platform that enables users to access enterprise applications through secure virtual desktops hosted in government data centers. The platform integrates **VDI, Zero Trust access, and multi-factor authentication**, providing a secure isolated workspace for accessing sensitive applications.

Key Capabilities

- Zero Trust Gateway for accessing VDI or Private applications
- Centralized virtual desktop provisioning and management
- Multi-Factor Authentication and Single Sign On
- Support for Windows and Linux virtual desktops
- Personal or shared desktop environments

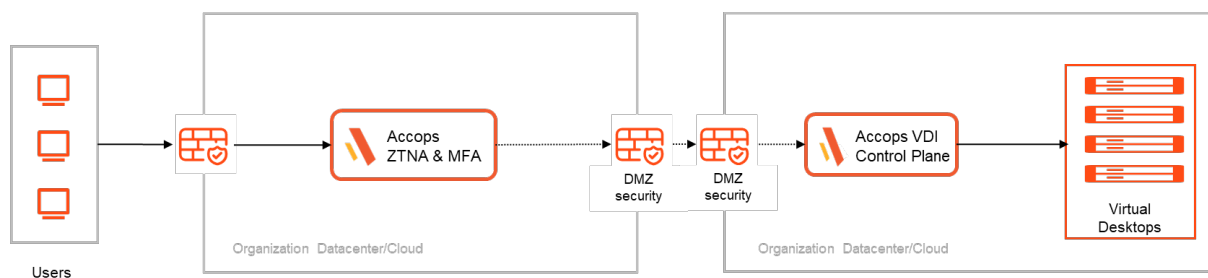
- Non-persistent virtual desktops for enhanced security
- Session monitoring and recording for audit compliance
- Posture checks, Device fingerprinting and geo-location validation
- Endpoint lockdown capabilities (USB blocking, screen capture prevention, data leak prevention)

Reference Architecture

The architecture consists of:

- **ZTNA Access Layer** – secure access gateway
- **Identity and MFA Integration** – user authentication layer
- **VDI Control Plane** – provisioning, orchestration and desktop brokering
- **Virtual Desktop Pools** – secure isolated desktops hosted in the data center

Users access virtual desktops through secure gateways using browsers, agents, or thin clients.



Use Cases

- **Securely Connect to All Corporate Resources through Centralized architecture:** Enable users to access any corporate application including web applications, client-server applications, RDP, SSH, and file servers through centrally hosted virtual desktops without exposing the internal network.
- **Provide Universal Access for All User Types:** Deliver secure, IT-controlled virtual desktop environments for internal employees, contractors, vendors, developers, auditors, and BYOD users, ensuring consistent security policies regardless of device or location.
- **Enable Secure Remote Work and Mobility:** Allow employees and field staff to securely access their virtual desktops from any location using laptops, thin clients, or browsers, while keeping applications and data within the data center environment.
- **Secure Development Environments for Developers:** Provide isolated virtual workspaces where developers can securely access code repositories, development tools, test environments, and application servers without exposing critical infrastructure or allowing source code to leave the data center.
- **Temporary Sandboxed Environments for External Developers:** Provision time-bound VDI environments for external development teams or vendors with restricted access, activity monitoring, and automatic workspace reset after session completion.
- **Secure Access for Auditors and Compliance Teams:** Provide auditors with controlled virtual desktops to review financial systems, audit logs, and compliance data without granting direct access to production systems or allowing sensitive data to be copied to local machines.
- **Isolated Environment for Privileged and Administrative Access:** Provide dedicated, isolated virtual desktops for administrators and privileged users to securely access critical infrastructure systems such as servers, databases, and management consoles with full activity monitoring and session recording.
- **Simplify Desktop Management and Security Compliance:** Centralize desktop provisioning, patch management, security policy enforcement, and monitoring through the VDI platform, reducing operational complexity while ensuring compliance with organizational security standards.
- **Enable Secure Thin Client Computing:** Replace traditional desktops with thin client devices connected to virtual desktops, reducing endpoint attack surface, simplifying device management, and improving overall infrastructure security.

Accops Remote Browser Isolation – Secure Internet Access

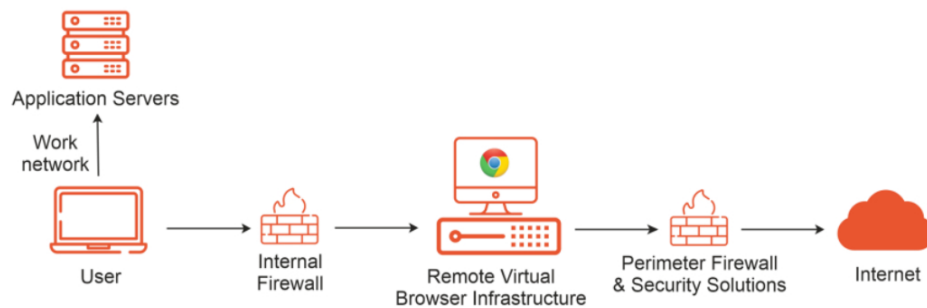
Product Overview

Accops Remote Browser Isolation provides a **secure virtual browser environment** where web browsing occurs on remote servers rather than on user endpoints. Only the browser display stream is transmitted to the user device, ensuring that web content never reaches the endpoint.

Key Capabilities

- Remote virtual browser execution
- Sandboxed browsing sessions
- Daily automated environment refresh
- Secure file downloads through scanning engines
- Prevention of browser plugin exploitation
- Integration with ZTNA secure access gateway
- Integration with AirBridge for secure file transfer

Reference Architecture



Use Cases

- Secure access to internet websites without exposing endpoint devices or internal networks to web-based threats.
- Safe browsing for users accessing untrusted or unknown websites during research or operational activities.
- Protection against malware, ransomware, and malicious scripts originating from web content.
- Secure internet access for users working on sensitive or classified systems.
- Controlled web access for developers, analysts, and research teams who need to browse external websites.

Accops Airbridge – Secure File Transfer Gateway

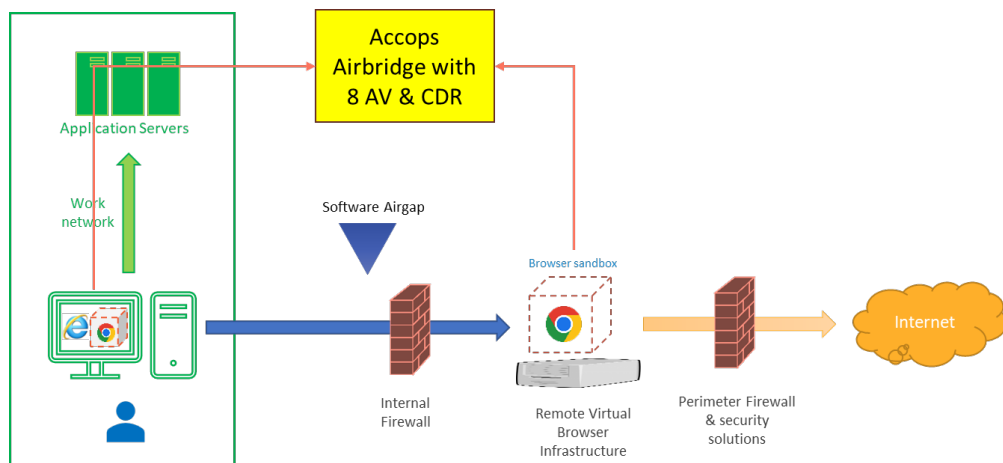
Product Overview

Accops Airbridge enables **secure file transfer between isolated networks**, such as between internet environments and internal networks. The solution ensures that files transferred across networks are thoroughly scanned and sanitized.

Key Capabilities

- Multi-layer malware scanning using multiple antivirus engines
- Content Disarm and Reconstruction (CDR) technology
- Controlled file upload and download workflows
- Whitelisted destination enforcement
- Secure transfer between internet and internal networks
- Role Based file download/upload

Reference Architecture



Use Cases

- Secure file transfer between internal systems and external users without exposing internal networks.
- Controlled exchange of sensitive documents with vendors, partners, and third-party organizations.
- Secure file movement between isolated or restricted network environments.
- Transfer of large files such as logs, reports, backups, and datasets in a controlled manner.
- Audited file sharing for compliance-sensitive environments.
- Secure upload and download of files for users accessing systems remotely.

Accops Huddle – Secure Remote Support Platform

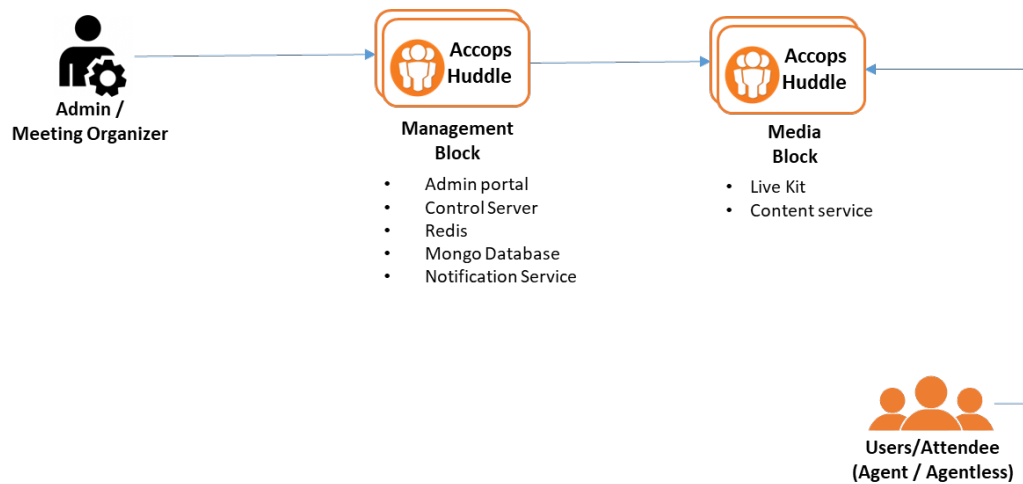
Product Overview

Accops Huddle is an on-premise remote meeting and troubleshooting platform designed for **IT helpdesk teams to securely support users across distributed environments**. The solution enables administrators to provide remote assistance without relying on internet-based remote desktop tools.

Key Capabilities

- Remote desktop sharing and control
- Air-Gapped & Isolated Network Support
- File transfer and chat (text, audio and video during support sessions)
- Multi-party collaboration sessions
- Session recording and detailed audit logs
- Role-based access control for administrators
- TLS encrypted communication
- Operation within isolated or air-gapped networks
- Clientless & Agent-Based Support

Reference Architecture



Use Cases

- Remote assistance for IT administrators to support servers, applications, and infrastructure without physical presence.
- Controlled vendor access for maintenance or troubleshooting activities.
- Joint troubleshooting sessions between multiple administrators when resolving complex issues.
- Secure support for branch or remote office IT environments.
- Supervised access for maintenance activities in sensitive or regulated systems.
- Knowledge transfer and guided support between senior engineers and operational teams

Accops HyDesk – Secure Thin Client Endpoint

Product Overview

Accops HyDesk devices are centrally managed thin client endpoints designed to replace traditional desktop PCs with secure and energy-efficient devices. These devices are optimized for accessing virtual desktops and private applications from centralized infrastructure.

Key Capabilities

- Centralized device management
- Secure kiosk mode operation
- Support for VDI environments
- Secure boot and hardened operating system
- Remote monitoring and troubleshooting
- USB device support and peripheral redirection
- Low power consumption and long hardware lifecycle

Reference Architecture



Feature	HY4000 family	HY6000 family
Hardware	Intel, 2 or 4 Cores, upto 8GB RAM, upto 128 GB SSD	Intel, 2 or 4 Cores, upto 8GB RAM, upto 1 TB SSD
OS	Accops HyDesk OS, Windows 10 IoT	Accops HyDesk OS, Windows 10 IoT/Pro
Ports	1 USB 3.0, 5 USB 2.0	3 USB 3.0, 1 Type C, 0-3 USB 2.0
	1 Gbps Ethernet, Optional Wi-Fi	1-2 Gbps Ethernet, Optional Wi-Fi
Display	1xDVI-D with VGA adaptor, 1xDP	4K, 1 HDMI, 1 additional VGA/DP/HDMI
Dual Monitor	Yes	Yes
Power	15W	15 W
Certifications	BIS, RoHS/REACH, UL, CC, IEC, EN, BIS	BIS, RoHS/REACH, UL, CC, IEC, EN, BIS
Warranty	Device 3 years, Power Adaptor 1 year	Device 3 years, Power Adaptor 1 year
CoO	China	Thailand / Taiwan

Use Cases

- Secure workstations for government employees
- Thin client computing for large offices
- Secure access terminals in controlled environments
- Cost-effective endpoint deployment across multiple offices